CS 4644/7643: Deep Learning
Fall 2026
Problem Set 4

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Discussions: https://piazza.com/class/ms43b2tjxa63lq#

Due: November 16, 2026, 11:59pm

Instructions

1. We will be using Gradescope to collect your assignments. Please read the following instructions for submitting to Gradescope carefully!

   - For the **HW4 Theory** component on Gradescope, upload one single PDF containing the answers to all the theory questions. **However, the solution to each problem/subproblem must be on a separate page. When submitting to Gradescope, please make sure to mark the page(s) corresponding to each problem/subproblem.** Questions with misassigned pages will be penalized by up to 1 point.
   - For the **HW4 Code** on Gradescope, please use the `collect_submission.py` script provided and upload **hw4_code_submission.zip**
   - For the **HW4 Notebook** component on Gradescope, please use the `collect_submission.py` script provided and upload the resulting **hw4_notebook_submission.pdf** on Gradescope.
   - Note: This is a large class and Gradescope's assignment segmentation features are essential. Failure to follow these instructions may result in parts of your assignment being penalized. We will not entertain regrading requests for failure to follow instructions.

2. LaTeX'd solutions are strongly encouraged, but scanned handwritten copies are acceptable. Hard copies are **not** accepted.

3. We generally encourage you to collaborate with other students.

   You may talk to a friend, discuss the questions and potential directions for solving them. However, you need to write your own solutions and code separately, and *not* as a group activity. Please list the students you collaborated with.

1 Collaborators [0.5 points]

Please list your collaborators and assign this list to the corresponding section of the outline on Gradescope. If you don't have any collaborators, please write 'None' and assign it to the corresponding section of the Gradescope submission regardless. Not assigning this question on Gradescope or leaving it blank will be marked as zero.

Solution

None

2 Variational Auto-Encoders

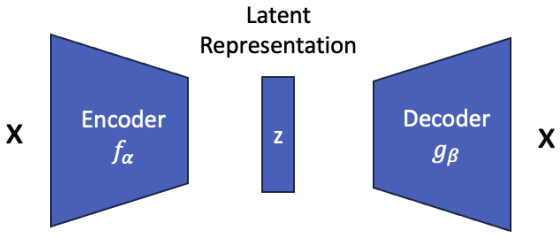


Figure 1: Standard auto-encoder that takes an input x, encodes it into a latent representation z, and then decodes back into the original x.

Variational Auto-Encoders (VAE) are latent variable models that can generate samples from a target distribution. Unlike standard autoencoders, VAEs learn a structured latent space that we can reliably sample from to generate new data. This problem walks you through deriving the ELBO training objective for VAEs.

1. **Understanding the ELBO [3 points + 1 point required for 7643/extra-credit for 4644]**

   Given a dataset X, we want to model the underlying distribution $p(x)$. A latent variable model assumes each observation x is generated from a lower-dimensional latent code z: first sample $z \sim p(z)$, then generate $x \sim p_\theta(x|z)$. Here $p(z)$ is a fixed prior (typically $\mathcal{N}(0, I)$) and $p_\theta(x|z)$ is a learned conditional distribution parameterized by θ. For our training objective, we wish to maximize the marginal log-likelihood:

   $$\log p_\theta(x) = \log \int p_\theta(x, z)\, dz = \log \int p_\theta(x|z)p(z)\, dz.$$

   However, computing this integral is intractable for deep latent variable models: there is no closed-form solution, and numerical integration over high-dimensional latent spaces is prohibitively expensive. To handle this intractability, we introduce an approximate posterior distribution $q_\phi(z|x)$ that we can tractably compute and sample from.

   (a) *Deriving the ELBO.* [2 pts]

   Derive a tractable lower bound on the log-likelihood by introducing the approximate posterior $q_\phi(z|x)$. Show that:

   $$\mathcal{L}(\theta, \phi; x) := \mathbb{E}_{z\sim q_\phi(z|x)}\left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)}\right] \leq \log p_\theta(x)$$

   This quantity $\mathcal{L}(\theta, \phi; x)$ is called the Evidence Lower Bound (ELBO). Any mathematically correct derivation is acceptable: Jensen's inequality, KL-divergence properties, etc. Some helpful identities:

   $$\log p_\theta(x) = \int q_\phi(z|x) \log p_\theta(x) dz = \mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x)] \qquad \frac{q_\phi(z|x)}{q_\phi(z|x)} = 1$$

Solution

We wish to maximize the marginal log-likelihood of the data:

$$\log p_\theta(x) = \log \int p_\theta(x, z)\, \mathrm{d}z$$

We introduce an arbitrary variational distribution $q_\phi(z \mid x)$ over the latent variable z such that $q_\phi(z \mid x) > 0$ whenever $p_\theta(x, z) > 0$. Multiplying and dividing the integrand by $q_\phi(z \mid x)$, we obtain:

$$\begin{aligned}\log p_\theta(x) &= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)}\, \mathrm{d}z \\ &= \log \mathbb{E}_{z \sim q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]\end{aligned}$$

Since $f(u) = \log(u)$ is a concave function, we apply Jensen's inequality, $\log \mathbb{E}[Y] \geq \mathbb{E}[\log Y]$:

$$\log \mathbb{E}_{z \sim q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right] \geq \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]$$

Defining the Evidence Lower Bound (ELBO) as:

$$\mathcal{L}(\theta, \phi; x) := \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]$$

Therefore:

$$\mathcal{L}(\theta, \phi; x) \leq \log p_\theta(x)$$

(b) *Decomposing the objective.* [1 pt]
Using the identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{KL}(q_\phi(z|x) \,\|\, p_\theta(z|x)),$$

explain why maximizing $\mathcal{L}(\theta, \phi; x)$ with respect to ϕ is equivalent to minimizing $D_{KL}(q_\phi(z|x) \,\|\, p_\theta(z|x))$. Why can't we minimize this KL directly instead of maximizing the ELBO?

Solution

We are given the identity:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$$

Equivalence of Maximizing ELBO and Minimizing KL Divergence: Rearranging the given identity yields:

$$\mathcal{L}(\theta, \phi; x) = \log p_\theta(x) - D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$$

Clearly, the true data marginal log-likelihood $\log p_\theta(x)$ is solely a function of the generative parameters θ and the data point x; it does not depend on the variational parameters ϕ. Consequently, with respect to ϕ, $\log p_\theta(x)$ acts as a constant. Maximizing $\mathcal{L}(\theta, \phi; x)$ with respect to ϕ directly minimizes $D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$, driving the variational distribution $q_\phi(z \mid x)$ closer to the true posterior $p_\theta(z \mid x)$.

Why Direct Minimization of KL is Intractable: The KL divergence between the approximate posterior and the true posterior is defined as:

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x)) = \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{q_\phi(z \mid x)}{p_\theta(z \mid x)} \right]$$

Evaluating this term requires computing the true posterior:

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)} = \frac{p_\theta(x \mid z) p(z)}{\int p_\theta(x \mid z) p(z) \, \mathrm{d}z}$$

The denominator $p_\theta(x)$ involves integrating over the high-dimensional latent space z, which is analytically intractable for deep neural networks. Because we cannot compute $p_\theta(z \mid x)$, we cannot evaluate or differentiate $D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$ directly. In contrast, the ELBO $\mathcal{L}(\theta, \phi; x)$ depends only on $p_\theta(x, z) = p_\theta(x \mid z) p(z)$ and $q_\phi(z \mid x)$, both of which are computable and differentiable.

(c) *Understanding the ELBO terms.* [1 pt] **(Required for 7643; Extra Credit for 4644)** From part (a), the ELBO can be rewritten as:

$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{KL}(q_\phi(z|x) \,\|\, p(z))}_{\text{regularization term}}$$

Suppose we train a VAE but remove the KL term entirely, optimizing only $\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)]$. What is one degenerate distribution $q_\phi(z|x)$ could collapse to? Why would this break generation at test time, even if reconstruction loss is low?

Solution

The decomposed ELBO is:

$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z))}_{\text{regularization term}}$$

Degenerate Distribution Without the KL Term: If the regularization term $D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z))$ is removed, the objective becomes purely the reconstruction loss:

$$\max_{\theta, \phi} \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)]$$

Under this unconstrained objective, the model acts as a standard deterministic autoencoder. The variance predicted by the encoder can collapse to zero. Alternatively, the encoder can map different inputs x to arbitrarily isolated, far-apart delta spikes in the latent space to eliminate any reconstruction ambiguity.

Why This Breaks Generation at Test Time: At test time, the generation procedure assumes the prior distribution:

$$z \sim p(z) = \mathcal{N}(0, I) \quad \text{followed by} \quad x \sim p_\theta(x \mid z)$$

Without the KL penalty enforcing the aggregate posterior $q(z) = \int q_\phi(z \mid x) p_{\mathrm{data}}(x) \, \mathrm{d}x$ to resemble the prior $p(z)$, the learned latent embeddings will occupy sparse, disconnected clusters with vast regions of empty, untrained space between them.

When sampling $z \sim \mathcal{N}(0, I)$ at test time, the sampled code will almost certainly land in an unmapped region that the decoder never observed during training. Consequently, the decoder outputs corrupted / blurry images, despite obtaining near-zero reconstruction error on the training set.

2. **Reparameterization Trick [1 point]**

To train the VAE using gradient descent, we need to compute gradients of the ELBO with respect to both θ and ϕ:

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \,\|\, p(z))$$

(a) *The reparameterization trick.* [1 pt]

For $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x)I)$, instead of sampling $z \sim q_\phi(z|x)$ directly, we:

i. Sample $\epsilon \sim \mathcal{N}(0, I)$

ii. Compute $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$

This gives us:

$$\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[\log p_\theta(x|\mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)]$$

Explain why this solves the gradient problem. Why can we now move ∇_ϕ inside the expectation?

Solution

To optimize the ELBO using gradient descent, we need to compute the gradient of the reconstruction term with respect to the encoder parameters ϕ:

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)] = \nabla_\phi \int q_\phi(z \mid x) \log p_\theta(x \mid z) \, \mathrm{d}z$$

Because the underlying sampling distribution $q_\phi(z \mid x)$ depends directly on ϕ, we cannot naively interchange the gradient operator ∇_ϕ and the expectation. Using the score-function estimator:

$$\nabla_\phi \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)] = \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log p_\theta(x \mid z) \nabla_\phi \log q_\phi(z \mid x)\right]$$

yields extremely high variance in Monte Carlo approximations, resulting in unstable or slow training.

The reparameterization trick expresses the random variable $z \sim \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x)I)$ as a deterministic, differentiable transformation of an independent base noise distribution:

$$z = g_\phi(\epsilon, x) = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \text{where } \epsilon \sim p(\epsilon) = \mathcal{N}(0, I)$$

Substituting this transformation into the expectation:

$$\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[\log p_\theta(x \mid \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)]$$

Because the distribution of $\epsilon \sim \mathcal{N}(0, I)$ has **no dependence on** ϕ, the measure under which the expectation is taken is fixed. Therefore, by the Leibniz integral rule, we can move the gradient operator ∇_ϕ inside the expectation:

$$\begin{aligned}
\nabla_\phi \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)] &= \nabla_\phi \int p(\epsilon) \log p_\theta(x \mid g_\phi(\epsilon, x)) \, \mathrm{d}\epsilon \\
&= \int p(\epsilon) \nabla_\phi \log p_\theta(x \mid g_\phi(\epsilon, x)) \, \mathrm{d}\epsilon \\
&= \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)} \left[\nabla_\phi \log p_\theta(x \mid g_\phi(\epsilon, x))\right]
\end{aligned}$$

Applying the chain rule gives:

$$\nabla_\phi \log p_\theta(x \mid g_\phi(\epsilon, x)) = \nabla_z \log p_\theta(x \mid z) \cdot \nabla_\phi g_\phi(\epsilon, x)$$

This pathwise derivative estimator has significantly lower variance and enables end-to-end backpropagation through both encoder and decoder networks.

3 Diffusion Models

1. Before starting section, we highly recommend reading the DDPM paper to familiarize yourself with the notation and equations.

   Denoising Diffusion Probabilistic Models (DDPMs) learn to reverse a gradual noising process. The forward process transforms data $x_0 \sim q(x_0)$ into noise through T steps, defined as a Markov chain:

   $$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

   where each transition adds Gaussian noise:

   $$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, \beta_t I).$$

   The goal is to learn a **reverse process** p_θ that inverts this noising:

   $$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$

   where $p(x_T) = \mathcal{N}(x_T; 0, I)$ and we model each reverse step as:

   $$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)).$$

   Starting from noise $x_T \sim \mathcal{N}(0, I)$ and iteratively sampling from $p_\theta(x_{t-1}|x_t)$, we can generate new data samples.

   The framework makes some key assumptions:

   (a) **Markov property**: $q(x_t|x_{t-1}, \ldots, x_0) = q(x_t|x_{t-1})$ — each step depends only on the previous timestep.

   (b) **Gaussian transitions with fixed schedule**: $\{\beta_t\}_{t=1}^{T}$ is predetermined with $0 < \beta_1 < \beta_2 < \cdots < \beta_T < 1$.

   (c) **Small noise per step**: When β_t is sufficiently small, the reverse process posterior $q(x_{t-1}|x_t)$ is approximately Gaussian (Sohl-Dickstein et al., 2015). This is why DDPMs model the reverse process $p_\theta(x_{t-1}|x_t)$ as a Gaussian.

2. **Forward diffusion [3 points]** Define the noise schedule $\{\beta_t\}_{t=1}^T$ with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. The forward process is

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1},\, \beta_t I).$$

Equation 4 in the DDPM paper gives a closed form expression for the forward process at an arbitrary timestep t:

$$q(x_t \mid x_0) = \mathcal{N}\Big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t) I\Big).$$

(a) *Closed form of $q(x_t|x_0)$* [2 pt]

Prove the expression above for $q(x_t|x_0)$ is true for all $t \geq 1$ by induction on t.

Helpful identities:

- *Reparametrization trick*: A random variable $X \sim \mathcal{N}(\mu, \sigma^2 I)$ can be written as $X = \mu + \sigma\epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$.
- *Sum of independent Gaussians*: If $X \sim \mathcal{N}(0, I)$ and $Y \sim \mathcal{N}(0, I)$ are independent, then $aX + bY \sim \mathcal{N}(0, (a^2 + b^2)I)$, or equivalently $aX + bY = \sqrt{a^2 + b^2}\,\epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$.

Solution

We want to prove by induction on t that for all $t \geq 1$:

$$q(x_t \mid x_0) = \mathcal{N}\Big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t) I\Big),$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

Base Case ($t = 1$): For $t = 1$, we have $\bar{\alpha}_1 = \alpha_1 = 1 - \beta_1$. By the definition of the forward step:

$$q(x_1 \mid x_0) = \mathcal{N}(\sqrt{\alpha_1}\, x_0,\, \beta_1 I) = \mathcal{N}(\sqrt{\bar{\alpha}_1}\, x_0,\, (1 - \bar{\alpha}_1) I).$$

Using the reparameterization trick, this can be written as:

$$x_1 = \sqrt{\alpha_1}\, x_0 + \sqrt{1 - \alpha_1}\, \epsilon_1 = \sqrt{\bar{\alpha}_1}\, x_0 + \sqrt{1 - \bar{\alpha}_1}\, \epsilon_1, \quad \text{where } \epsilon_1 \sim \mathcal{N}(0, I).$$

Thus, the base case holds.

Inductive Hypothesis: Assume the statement holds for some $t - 1 \geq 1$:

$$q(x_{t-1} \mid x_0) = \mathcal{N}\Big(\sqrt{\bar{\alpha}_{t-1}}\, x_0,\ (1 - \bar{\alpha}_{t-1}) I\Big),$$

which means we can write:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\, \epsilon^*_{t-1}, \quad \text{where } \epsilon^*_{t-1} \sim \mathcal{N}(0, I).$$

Inductive Step: By definition, the transition from x_{t-1} to x_t is $q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1}, \beta_t I)$ with $\beta_t = 1 - \alpha_t$. Reparameterizing this step:

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_t, \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I),$$

with ϵ_t independent of ϵ^*_{t-1}. Substituting the inductive hypothesis for x_{t-1}:

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\left(\sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1-\bar{\alpha}_{t-1}}\, \epsilon^*_{t-1}\right) + \sqrt{1-\alpha_t}\, \epsilon_t \\ &= \sqrt{\alpha_t \bar{\alpha}_{t-1}}\, x_0 + \sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}\, \epsilon^*_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_t. \end{aligned}$$

Since $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$, the deterministic mean term is $\sqrt{\bar{\alpha}_t}\, x_0$.
For the stochastic terms, using the sum of independent gaussians trick, which says the linear combination of two independent standard Gaussians $aX + bY$ (with $X, Y \sim \mathcal{N}(0, I)$) is distributed as $\sqrt{a^2+b^2}\, \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. Setting $a = \sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}$ and $b = \sqrt{1-\alpha_t}$:

$$\begin{aligned} a^2 + b^2 &= \alpha_t(1-\bar{\alpha}_{t-1}) + (1-\alpha_t) \\ &= \alpha_t - \alpha_t \bar{\alpha}_{t-1} + 1 - \alpha_t \\ &= 1 - \alpha_t \bar{\alpha}_{t-1} \\ &= 1 - \bar{\alpha}_t. \end{aligned}$$

Therefore:

$$\sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}\, \epsilon^*_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_t = \sqrt{1-\bar{\alpha}_t}\, \bar{\epsilon}_t, \quad \text{where } \bar{\epsilon}_t \sim \mathcal{N}(0, I).$$

Substituting this back yields:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \bar{\epsilon}_t \implies q(x_t \mid x_0) = \mathcal{N}\Big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I\Big).$$

By induction, the closed-form expression holds for all $t \geq 1$.

(b) *Noise distribution* $q(x_T|x_0)$ [1 pt]

In diffusion models, we choose $\{\beta_t\}$ such that $\bar{\alpha}_T \approx 0$. What is $q(x_T|x_0)$ when $\bar{\alpha}_T = 0$? Why is this useful for generation?

Solution

Substituting $\bar{\alpha}_T = 0$ into the closed-form expression $q(x_T \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_T}\, x_0, (1 - \bar{\alpha}_T)I)$:

$$q(x_T \mid x_0) = \mathcal{N}(\mathbf{0}, I).$$

This is useful because when $\bar{\alpha}_T \approx 0$, all statistical dependency on the initial data point x_0 vanishes, and the distribution of latent variables at the final timestep T converges completely to pure standard isotropic Gaussian noise $\mathcal{N}(0, I)$, regardless of the input data distribution.

This makes ancestral sampling tractable at test time: we can start the generation process simply by drawing a random sample $x_T \sim \mathcal{N}(0, I)$ without needing access to any ground-truth data, and then iteratively apply the learned reverse transitions $p_\theta(x_{t-1} \mid x_t)$ down to $t = 0$ to synthesize new images.

3. **Reverse Process [2 points + 1 point required for 7643/extra-credit for 4644]**

(a) *KL divergence between Gaussians* [1 pt]
Recall that the PDF of a univariate Gaussian $\mathcal{N}(\mu, \sigma^2)$ is:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

and the KL divergence between two distributions p and q is:

$$D_{KL}(p\|q) = \mathbb{E}_{x\sim p}\left[\log \frac{p(x)}{q(x)}\right]$$

Show that when $p = \mathcal{N}(\mu_1, \sigma^2)$ and $q = \mathcal{N}(\mu_2, \sigma^2)$ share the same variance:

$$D_{KL}(\mathcal{N}(\mu_1, \sigma^2)\|\mathcal{N}(\mu_2, \sigma^2)) = \frac{1}{2\sigma^2}(\mu_1 - \mu_2)^2$$

Solution

Let $p(x) = \mathcal{N}(x; \mu_1, \sigma^2)$ and $q(x) = \mathcal{N}(x; \mu_2, \sigma^2)$ be two univariate Gaussians with equal variance σ^2. Their probability density functions are:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right), \qquad q(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right).$$

The log-ratio of the PDFs is:

$$\begin{aligned}
\log \frac{p(x)}{q(x)} &= \log p(x) - \log q(x) \\
&= \left(-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x-\mu_1)^2}{2\sigma^2}\right) - \left(-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x-\mu_2)^2}{2\sigma^2}\right) \\
&= \frac{1}{2\sigma^2}\left[(x-\mu_2)^2 - (x-\mu_1)^2\right].
\end{aligned}$$

$$\begin{aligned}
(x-\mu_2)^2 - (x-\mu_1)^2 &= (x^2 - 2x\mu_2 + \mu_2^2) - (x^2 - 2x\mu_1 + \mu_1^2) \\
&= 2x(\mu_1 - \mu_2) + (\mu_2^2 - \mu_1^2).
\end{aligned}$$

Taking the expectation under $x \sim p = \mathcal{N}(\mu_1, \sigma^2)$, we use $\mathbb{E}_{x\sim p}[x] = \mu_1$:

$$\begin{aligned}
D_{\mathrm{KL}}(p\,\|\,q) &= \mathbb{E}_{x\sim p}\left[\log \frac{p(x)}{q(x)}\right] \\
&= \frac{1}{2\sigma^2}\mathbb{E}_{x\sim p}\left[2x(\mu_1 - \mu_2) + (\mu_2^2 - \mu_1^2)\right] \\
&= \frac{1}{2\sigma^2}\left[2\mu_1(\mu_1 - \mu_2) + \mu_2^2 - \mu_1^2\right] \\
&= \frac{1}{2\sigma^2}\left[2\mu_1^2 - 2\mu_1\mu_2 + \mu_2^2 - \mu_1^2\right] \\
&= \frac{1}{2\sigma^2}\left[\mu_1^2 - 2\mu_1\mu_2 + \mu_2^2\right] \\
&= \frac{1}{2\sigma^2}(\mu_1 - \mu_2)^2.
\end{aligned}$$

(b) *Connecting to DDPM* [1 pt]

Using the result from part (a), explain in 1-2 sentences why $L_{t-1} = D_{KL}(q(x_{t-1}|x_t, x_0)\|p_\theta(x_{t-1}|x_t))$ simplifies to:

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C$$

What key assumption on p_θ makes part (a) applicable?

Solution

Both $q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$ and the model reverse distribution $p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I)$ are isotropic Gaussians. Under the shared diagonal covariance assumption, the KL divergence factorizes into a sum over coordinate-wise 1D Gaussian KL divergences, yielding $\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2$ plus a constant term C independent of θ (arising from any fixed ratio of variances). Taking the expectation over the forward trajectories produces the stated objective:

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C.$$

The key assumption is that the reverse covariance is fixed to an isotropic scalar matrix that is not learned by θ:

$$\Sigma_\theta(x_t, t) = \sigma_t^2 I,$$

where σ_t^2 is fixed to either β_t or $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$. Because the covariance matrices are fixed and proportional to the identity, the KL divergence reduces strictly to the squared Euclidean distance between the mean vectors.

(c) *Forward process posterior* [1 point] **(Required for 7643; Extra Credit for 4644)**: In Appendix A, the authors rewrite the $q(x_{t-1}|x_t)$ terms as $q(x_{t-1}|x_t, x_0)$. Why can't we compute $q(x_{t-1}|x_t)$? Why is $q(x_{t-1}|x_t, x_0)$ a valid substitution?

Solution

We cannot compute $q(x_{t-1} \mid x_t)$ because by Bayes' rule, the reverse conditional distribution of the forward process is:

$$q(x_{t-1} \mid x_t) = \frac{q(x_t \mid x_{t-1})q(x_{t-1})}{q(x_t)} = \frac{q(x_t \mid x_{t-1}) \int q(x_{t-1} \mid x_0)q(x_0)\,\mathrm{d}x_0}{\int q(x_t \mid x_0)q(x_0)\,\mathrm{d}x_0}.$$

Evaluating this density requires integrating over an unknown, high-dimensional empirical data distribution $q(x_0)$, which is analytically intractable. Hence, $q(x_{t-1} \mid x_t)$ cannot be evaluated in closed form.

Conditioning on x_0 to use $q(x_{t-1} \mid x_t, x_0)$ is a valid substitution:

i. When conditioned on x_0, all participating terms in Bayes' theorem are tractable Gaussians:

$$q(x_{t-1} \mid x_t, x_0) = q(x_t \mid x_{t-1}, x_0)\frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} = q(x_t \mid x_{t-1})\frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}.$$

Because the forward transitions and marginals conditioned on x_0 are known Gaussians ($q(x_t \mid x_0)$ and $q(x_{t-1} \mid x_0)$), standard Gaussian conjugate multiplication yields a closed-form Gaussian distribution $\mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$.

ii. The training objective optimizes the expected variational bound across the entire data distribution:

$$\mathbb{E}_{q(x_{0:T})}\left[\log \frac{q(x_{1:T} \mid x_0)}{p_\theta(x_{0:T})}\right].$$

During training, clean training samples $x_0 \sim q(x_0)$ are always observed and available. By applying the chain rule of probabilities and the Markov property:

$$q(x_{1:T} \mid x_0) = q(x_T \mid x_0) \prod_{t=2}^{T} q(x_{t-1} \mid x_t, x_0),$$

we can algebraically expand the trajectory joint ratio into individual KL divergence terms conditioned on x_0 without altering the expectation over $q(x_0)$.

4 Paper Review [2 Points]

For this homework, you'll be writing a paper review for **Denoising Diffusion Probabilistic Models**. We recommend reading this paper before you start the theory and coding sections on diffusion models.
The paper introduces a method for high quality image generation inspired by non-equilibrium thermodynamics. Diffusion models operate through a forward noising process and a reverse denoising process. These state of the art generative models are useful in many domains including image generation and robotics. The paper can be viewed here.

The review will be graded using the following evaluation rubric:

1. **[1 point]** Briefly summarize the key contributions, strengths and weaknesses of this paper.

Solution

The central idea of DDPM is simple: instead of forcing a network to synthesize complex images in one leap, it breaks the task into hundreds of tiny, manageable steps. By defining a forward process akin to Brownian motion, where structured data gradually diffuses into isotropic Gaussian noise, the paper reframes image generation as learning the reverse drift that nudges noise back toward the data distribution. Because the network is trained across arbitrary noise levels rather than fitting discrete training points directly, it learns the smooth probability density of the whole manifold, theoretically reducing the possibliity of the model simply memorizing training examples. Their core contribution is reparameterizing this reverse process so the network simply predicts the noise added at each step, optimized via an elegant, unweighted mean squared error objective. The main strength is exceptional training stability and visual fidelity that rivals GANs without mode collapse or adversarial instability. The primary bottleneck remains sampling speed: generating a single image requires evaluating a large U-Net hundreds of times sequentially, making inference far slower than single-pass generators.

2. **[1 point]** What is your personal takeaway from this paper?

Solution

My biggest takeaway is the authors' departure from strict likelihood theory to achieve perceptual realism. The formal variational bound mathematically insists on heavily penalizing errors at very small noise levels, causing models to spend their capacity resolving imperceptible, pixel-level grain. By dropping these theoretical weighting factors, the authors created a simplified loss that naturally shifts the model's focus to intermediate steps where global structure and semantics take shape. Its also a strong claim to the idea that the objective that may yield the tightest bound mathematically isn't necessarily the one that produces the best perceptual results. Bridging non-equilibrium physics with DL turned a hard problem: modeling high-dimensional probability densities, into an iterative regression task.

Guidelines: Please restrict your reviews to at least one paragraph per question, but no more than 350 words (total length for answers to both of the above questions). Please write separate answers for each question to assist in Gradescope grading.